
ECE 6553 - OPTIMAL CONTROLS

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1 Reachability of Linear Systems

1.1 Reachability of an LTI System

Consider the Linear Time-Invariant (LTI) finite system

$$\begin{aligned}\dot{x} &= Ax + bu \\ y &= cx\end{aligned}$$

with $A \in \mathbb{R}^{n \times n}$, $b \in \mathbb{R}^{n \times 1}$, $c \in \mathbb{R}^{1 \times n}$. We define our system to be $\Sigma = (A, b, c)$. The reachability matrix $\mathcal{R}$ is

$$\mathcal{R}(A, b) = \begin{bmatrix} b & Ab & \dots & A^{n-1}b \end{bmatrix}$$

We determine reachability for the system by looking to see if $\det \mathcal{R}(A, b) \neq 0$. In other words, if the reachability matrix is invertible (non-singular), then we have reachability. Reachability means that given some initial state, the control system can drive it to any other state within its state space.

Say we are given $u(t)$ and $y(t)$ for $t > 0$. Can we find $x(0)$? The observability matrix $\mathcal{O}$ is

$$\mathcal{O}(A, c) = \begin{bmatrix} c \\ cA \\ \vdots \\ cA^{n-1} \end{bmatrix}$$

The system is observable if $\det \mathcal{O}(A, c) \neq 0$, which means we would be able to determine $x(0)$. observability means that the system's internal state can be fully understood with only the external outputs.

However, there is an easier way to determine reachability without computing the reachability matrix. The Popov-Belerith-Hautus (PBH) test says that for the augmented matrix

$$\begin{bmatrix} sI - A & | & b \end{bmatrix} \in \mathbb{R}^{n \times (n+1)}$$

that we can determine reachability if

$$\text{rank} \begin{bmatrix} sI - A & | & b \end{bmatrix} = n, \forall s$$

1.2 Fundamental Realization Theorem

We can realize a transfer function for an LTI system $\Sigma = (A, b, c)$ if (A, b) is reachable and (A, c) is observable. This leads to the realization of

$$H(s) = \frac{b(s)}{a(s)} = \frac{c \text{Adi}(sI - A)b}{\det(sI - A)}$$

The adioint (or adjugate) matrix for a matrix A is defined as the matrix that satisfies

$$A^{-1} = \frac{1}{\det(A)} \text{Adi}(A)$$

and is also the transpose of the cofactor matrix of A . The cofactor matrix C of A is defined to have the following elements:

$$C_{ii} = (-1)^{i+i} \det(M_{ii})$$

where M_{ii} is the minor matrix created by removing the i -th row and i -th column of A . So then the adjoint matrix would be

$$\text{Adj}(A)_{ii} = C_{ii} = (-1)^{i+i} \det(M_{ii})$$

The *minimal* realization is when $a(s)$ and $b(s)$ are coprime or relatively prime.

1.3 Linear Time-Variant Systems

We have looked at time-invariant systems, but what if the behavior of the system also depends on time? Consider the scalar system

$$\dot{x}(t) = a(t)x(t), \quad x(0) = x_0$$

We see that

$$\begin{aligned} \frac{\dot{x}(t)}{x(t)} &= \frac{d}{dt} \ln x(t) = a(t) \\ \ln x(t) - \ln x(0) &= \int_0^t a(\tau) d\tau \\ \ln x(t) &= \ln x(0) + \int_0^t a(\tau) d\tau \\ x(t) &= x(0) \exp \int_0^t a(\tau) d\tau \end{aligned}$$

However, this is only true for the scalar case. For the vector case, the solution is more complex:

$$x(t) = x_0 + x_0 \int_0^t A(\tau) d\tau + x_0 \int_0^t A(\tau) \int_0^\tau A(\tau_1) d\tau_1 d\tau + \dots$$

Then, for a non-zero initial value, we end up with:

$$\begin{aligned} x(t) &= x(\tau) + x(\tau) \int_\tau^t A(\theta) d\theta + \dots \\ &= \phi(t, \tau) x(\tau) \end{aligned}$$

where $\phi(t, \tau)$ is our transition matrix from time τ to time t . Now, consider the system

$$\dot{x}(t) = A(t)x(t) + b(t)u(t)$$

The solution is then

$$x(t) = \phi(t, \tau) x(\tau) + \int_\tau^t \phi(t, \theta) b(\theta) u(\theta) d\theta$$

1.4 Instantaneous Reachability

Consider the system

$$\dot{x}(t) = A(t)x(t) + b(t)\delta(t)$$

which has impulsive input. We know that $\delta(t) = H'(t)$, so

$$\begin{aligned}\dot{x}(t) &= A(t)x(t) + b(t)\frac{d}{dt}H(t) \\ &= A(t)x(t) + \frac{d}{dt}[b(t)H(t)] - H(t)\frac{d}{dt}b(t) \quad (\text{by product rule})\end{aligned}$$

$$\begin{aligned}\frac{d}{dt}[x(t) - b(t)H(t)] &= A(t)x(t) - H(t)\frac{d}{dt}b(t) \\ &= A(t)[x(t) - b(t)H(t)] + H(t)[A(t)b(t) - \dot{b}(t)]\end{aligned}$$

This is of the form

$$\frac{d}{dt}z(t) = A(t)z(t) + \text{discontinuous function}$$

where z is continuous.

From further derivation, we can see that for the system

$$\dot{x}(t) = A(t)x(t) + b(t)\delta^{(k)}(t)$$

that

$$x(0+) = ([A(t) - D]^k b(t))_{t=0} + x(0-)$$

where the operator $(\cdot)_{t=0}$ means evaluated at $t = 0$ and D is the derivative operator such that $Df = \dot{f}$. Consider now the input

$$u(t) = \sum_{i=0}^{N-1} \alpha_i \delta^{(i)},$$

we get that

$$x(0+) = x(0-) + [b, (A - D)b, (A - D)^2b, \dots] \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \vdots \end{bmatrix}$$

So, the infinite reachability matrix is

$$\mathcal{R}(A, b) = [b, (A - D)b, (A - D)^2b, \dots]$$

Note that the entries are functions dependent on time t , so the time-variant reachability also depends on t .

1.5 Finite Time Reachability

For $t \rightarrow T$,

$$x_f = x(T) = \phi(T, \tau)x(\tau) + \int_{\tau}^T \phi(T, \theta)b(\theta)u(\theta)d\theta$$

Then, let $u(\theta)$ be of the form

$$u(\theta) = b(\theta)^{\top} \phi^{\top}(T, \theta)q$$

where q is some unknown constant vector. Our solution becomes

$$x(T) = \phi(T, \tau)x(\tau) + q \int_{\tau}^T \phi(T, \theta)b(\theta)b(\theta)^{\top} \phi^{\top}(T, \theta)d\theta$$

We call this the reachability Gramian, defined as

$$\mathcal{R}_{(t_0, t_f)}(A, b) = \int_{t_0}^{t_f} \phi(t_f, t)b(t)b(t)^{\top} \phi^{\top}(t_f, t)dt$$

The optimal input $u^*(t)$ is the input that will drive our state from $(x_0, t_0) \rightarrow (x_f, t_f)$ where $t_f > t_0$. We define it as such:

$$u^*(t) = b^{\top}(t)\phi^{\top}(t_f, t) [\mathcal{R}_{(t_0, t_f)}(A, b)]^{-1} (x_f - \phi(t_f, t_0)x_0)$$

A unique property of this optimal input is that it is the input with minimal energy:

$$u^*(t) = \min_u \mathcal{E}_u$$

$$\mathcal{E}_u = \frac{1}{2} \int_{t_0}^{t_f} u^2(t)dt$$

2 Linearization of a Nonlinear System

Consider the system

$$\dot{x} = f(x, u)$$

$$y = h(x, u)$$

$$x \in \mathbb{R}^n, f = \begin{bmatrix} f_1 \\ \vdots \\ f_n \end{bmatrix}$$

We say that x_e is an equilibrium point if $f(x_e, 0) = 0$. Stability of nonlinear systems is defined by the stability of the equilibrium points, not of the system itself. But how do we find if an equilibrium point is stable or not (i.e., look at the behavior of $\dot{x}$ around x_e)?

We perturb the equilibrium point by a small amount $\tilde{x}$: $x = x_e + \tilde{x}$. Then,

$$\dot{x} = \dot{x}_e + \dot{\tilde{x}} = f(x_e + \tilde{x}, 0)$$

Since $\dot{x}_e$ is a constant, the derivative is 0. So,

$$\dot{\tilde{x}} = f(x_e + \tilde{x}, 0) = f(x_e, 0) + \left. \frac{\partial f}{\partial x} \right|_{x_e, 0} \tilde{x} + \text{higher order terms}$$

where

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

is the Jacobian. Since $\tilde{x}$ is small, we can rewrite our system as

$$\dot{\xi} = \left. \frac{\partial f}{\partial x} \right|_{(x_e, 0)} \xi$$

with $\xi \approx \tilde{x}$. We see that this is a linear time-invariant system. This is an example of linearizing a nonlinear system.

Let us look at the behavior of the system around the initial values (x_0, u_0) . We perturb x_0 and u_0 :

$$\begin{aligned} x &= x_0 + \tilde{x} \\ u &= u_0 + \tilde{u} \end{aligned}$$

So,

$$\begin{aligned} \dot{x} &= \dot{x}_0 + \text{tilde} \dot{x} = f(x_0 + \tilde{x}, u_0 + \tilde{u}) \\ \dot{\tilde{x}} &= f(x_0, u_0) + \left. \frac{\partial f}{\partial x} \right|_{(x_0, u_0)} \tilde{x} + \left. \frac{\partial f}{\partial u} \right|_{(x_0, u_0)} \tilde{u} + \text{h.o.t.} \end{aligned}$$

Then our linearized system with $\xi \approx \tilde{x}$ is

$$\dot{\xi} = \left. \frac{\partial f}{\partial x} \right|_{(x_0(t), u_0(t))} \xi + \left. \frac{\partial f}{\partial u} \right|_{(x_0(t), u_0(t))} \tilde{u}$$

which is a linear time-variant system.

2.1 Stability of the Equilibrium

Consider the equilibrium point (x_e, u_e) . For input $u(t) = u_e$, we say that the equilibrium point is stable if the solution $x(t)$ converges to x_e . Let the Jacobian $\left. \frac{\partial f}{\partial x} \right|_{(x_e, u_e)}$ be the matrix A . The linearized system is $\dot{\xi} = A\xi$.

If $\text{Re}\lambda(A) < 0$, then (x_e, u_e) is locally asymptotically stable.

If there is an eigenvalue of A with a positive real part, then the equilibrium is locally unstable.

If an eigenvalue of A is purely imaginary, then no conclusion can be drawn.

2.1.1 Hartman-Grossman Theorem

The equilibrium (x_e, u_e) for $\dot{x} = f(x, u)$ is said to be *hyperbolic* if $\text{Re}\lambda(A) \neq 0$ where $A = \left. \frac{\partial f}{\partial x} \right|_{(x_e, u_e)}$.

Theorem 2.1. If (x_e, u_e) is a hyperbolic equilibrium, then the global stability of the linearized model implies the local stability of the equilibrium for the nonlinear system.

3 Parameter Optimization

Given

$$L(x) \in \mathbb{R}, \quad x \in \mathbb{R}^n$$

we want to find the optimal parameter x^* such that $L(x^*)$ is an extremum. The things we have to investigate to do this are:

1. Existence: “Does an extremum even exist?”
2. Uniqueness: Guarantee that different methods converge to the same solution
3. Sufficient conditions: If they are satisfied, we can say we have an extremum
4. Necessary conditions: Must be satisfied but do not guarantee an extremum
5. Computation methods

3.1 Computational Methods

3.1.1 Geometric Methods

3.1.2 Algebraic Methods

The algebraic-geometric mean inequality says that for $x_i > 0$ for $i = 1, \dots, n$, the arithmetic mean (AM) is

$$\text{AM}\{x_i\} = \frac{1}{n} \sum_{i=1}^n x_i$$

and the geometric mean (GM) is

$$\text{GM}\{x_i\} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}$$

which follow the following inequality:

$$\text{AM}\{x_i\} \geq \text{GM}\{x_i\}$$

The quadratic mean (QM) is

$$\text{QM}\{x_i\} = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}$$

and then $\text{AM}\{x_i\} \leq \text{QM}\{x_i\}$. Finally, the harmonic mean (HM) is

$$\text{HM}\{x_i\} = \left[\frac{1}{n} \sum_{i=1}^n \frac{1}{x_i} \right]^{-1}$$

It holds that

$$0 < \text{HM} \leq \text{GM} \leq \text{AM} \leq \text{QM}$$

Matrix Inequalities:

Let $A \in \mathbb{R}^{n \times n}$ be a matrix. Then let

$$A_s = \frac{A + A^\top}{2}$$

$$A_A = \frac{A - A^\top}{2}$$

then, $A_s^\top = A_s$ and $A_A^\top = -A_A$.

Now we look at the quadratic form. Let P be a symmetric matrix.

$$x^\top P x, \quad x \in \mathbb{R}^n$$

Note the properties:

$$x^\top A x = x^\top A_s x$$

$$x^\top A_A x = 0$$

Consider:

$$\min_{\|x\|=1} x^\top P x = ?$$

Recall that

$$P = V \Lambda V^{-1} = V \Lambda V^\top$$

Then,

$$x^\top P x = x^\top V \Lambda V^\top x$$

Let $y = V^\top x$:

$$x^\top V \Lambda V^\top x = y^\top \Lambda y = \sum_{i=1}^n y_i^2 \lambda_i$$

Note that

$$\|y\|^2 = y^\top y = x^\top V V^\top x = x^\top x = \|x\|^2$$

so length does not change. Therefore if $\|x\| = 1$, then $\|y\| = 1$. So,

$$\min_{\|x\|=1} x^\top P x = \min_{\|y\|=1} y^\top \Lambda y = \text{minimum eigenvalue of } P$$

Theorem 3.1. Courant-Fisher Theorem

Consider that $x \perp \text{span}\{w_1, \dots, w_{n-k}\}$, then

$$\max_{\|x\|=1} x^\top P x = \lambda_k$$

3.1.3 Analytic Methods

Theorem 3.2. Extreme Value Theorem

If f is continuous with mapping from a compact (closed and bounded) set to $\mathbb{R}$, then f will always have a maximum and a minimum.

A counterexample is seen by considering $f(t) = e^t$ in the domain (a, b) where $-\infty < a < b < \infty$. Then $e^a < e^t, \forall t \in (a, b)$. We say that e^a is the infimum of $\{e^t\}$ in (a, b) .

Weistrass's Theorem:

For the 1-D case:

1. Consider the necessary conditions for extremum in $[a, b]$
2. The sufficient conditions for a local minimum at $x^* \in [a, b]$ are:

$$f'(x^*) = 0$$

$$f''(x^*) > 0 \text{ if } x^* \text{ is in the interior } (a, b)$$

$$\text{if } x^* = a \rightarrow f'(x^*) \geq 0, f''(x^*) > 0$$

3. Existence: we see that if f is continuous in $[a, b]$, then there must be a minimum
4. Unique if f is strictly convex, it has a minimum at a unique point in $[a, b]$

Derivatives with respect to Matrices:

Consider

$$L : \mathbb{R}^{n \times m} \rightarrow \mathbb{R}$$

$$L(X) = \text{Tr}(AX)$$

where Tr is the trace function:

$$\text{Tr}(M) = \sum_{i=1}^n M_{ii}$$

with properties

$$\text{Tr}(M^\top) = \text{Tr}(M)$$

$$\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB)$$

Then,

$$\text{Tr}(AX) = \sum_{i=1}^n (AX)_{ii} = \sum_{i=1}^m \sum_{i=1}^n A_{ii} X_{ii}$$

The derivative with respect to X is a matrix with elements

$$\left(\frac{\partial L}{\partial X} \right)_{ii} = \frac{\partial L}{\partial X_{ii}}$$

So,

$$\begin{aligned}
\left(\frac{\partial}{\partial X}\text{Tr}(AX)\right)_{ii} &= \frac{\partial}{\partial X_{ii}}\text{Tr}(AX) \\
&= \frac{\partial}{\partial X_{ii}} \sum_{\alpha=1}^m \sum_{\beta=1}^n A_{\alpha\beta} X_{\beta\alpha} \\
&= \sum_{\alpha=1}^m \sum_{\beta=1}^n A_{\alpha\beta} \frac{\partial X_{\beta\alpha}}{\partial X_{ii}}
\end{aligned}$$

We then introduce the Kronecker Delta:

$$\delta_{\alpha\beta} = \begin{cases} 1, & \alpha = \beta \\ 0, & \text{otherwise} \end{cases}$$

Then,

$$\begin{aligned}
\left(\frac{\partial}{\partial X}\text{Tr}(AX)\right)_{ii} &= \sum_{\alpha=1}^m \sum_{\beta=1}^n A_{\alpha\beta} \delta_{\beta i} \delta_{\alpha i} \\
&= \sum_{\beta=1}^n A_{i\beta} \delta_{b\beta i} \\
&= A_{ii}
\end{aligned}$$

3.1.4 Numerical Methods

3.2 Parameter Optimization with Equality Constraints

We define a problem with equality constraints to be one where we want to find an extremum for $L(x)$ with constraint $f(x) = 0$. For example, say there is a constraint that $x_1 = x_2$. Then, $f(x) = x_1 - x_2 = 0$ is our constraint.

Let $L : \mathbb{R}^n \rightarrow \mathbb{R}$ be the function to be minimized. We want to find x^* such that $L(x^*)$ is minimal and $f(x^*) = 0$ with $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $m < n$. We define x to be

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \\ u_1 \\ \vdots \\ u_{n-m} \end{bmatrix}$$

Then,

$$\begin{aligned}
dL(x) &= \partial L(y, u) = \frac{\partial L}{\partial y} dy + \frac{\partial L}{\partial u} du \\
&= \left(\frac{\partial L}{\partial u} - \frac{\partial L}{\partial y} \left(\frac{\partial f}{\partial y} \right)^{-1} \frac{\partial f}{\partial u} \right) du
\end{aligned}$$

We want this to be 0. We define

$$\lambda^\top = -\frac{\partial L}{\partial y} \left(\frac{\partial f}{\partial y} \right)^{-1}$$

These are the Lagrange multipliers. Our optimization goal becomes

$$\frac{\partial L}{\partial u} + \lambda^\top \frac{\partial f}{\partial u} = 0$$

We define the Hamiltonian H to be

$$H = L + \lambda^\top f$$

so that the optimization target (sufficient conditions) is

$$\frac{\partial H}{\partial u} = 0, \frac{\partial H}{\partial \lambda} = 0$$

With the necessary conditions

$$\frac{\partial H}{\partial x} = 0 \rightarrow \frac{\partial H}{\partial y} = 0, \frac{\partial H}{\partial u} = 0$$

Example Problem:

Let $L(x) = x_1^2 - x_2$ with the constraint $x_1 = x_2 + 1$.

Then, $f(x) = x_1 - x_2 - 1$. So, our Hamiltonian is

$$H = L + \lambda^\top f = x_1^2 - x_2 + \lambda(x_1 - x_2 - 1)$$

The necessary conditions are

$$\frac{\partial H}{\partial x_1} = 2x_1 + \lambda = 0 \rightarrow x_1 = -\frac{\lambda}{2}$$

$$\frac{\partial H}{\partial x_2} = -1 - \lambda = 0 \rightarrow \lambda = -1$$

So, $x_1 = \frac{1}{2}$. Then, we look at our constraint:

$$\frac{\partial H}{\partial \lambda} = x_1 - x_2 - 1 = 0 \rightarrow x_2 = x_1 - 1$$

So, $x_2 = -\frac{1}{2}$. Then, our optimal solution is

$$x^* = \left(\frac{1}{2}, -\frac{1}{2} \right)$$

3.3 Parameter Optimization with Inequality Constraints

Instead of $f(x) = 0$, our problem becomes finding an extremum x^* of $L(x)$ such that $f(x^*) \leq 0$. Our Hamiltonian is

$$H = L + \mu^\top f$$

where $\mu \geq 0$. For this, we have to note the idea of inactive and active constraints. A constraint is inactive if the extremum of the function already satisfies $f(x) \leq 0$. A constraint is active if the extremum lies in $f(x) > 0$ so we have to find the next smallest/largest value that does satisfy $f(x) \leq 0$. When solving for a parameter optimization problem like this, we look at each possible combination of inactive and active constraints. For example, if there are two constraints, we look at the cases where (a) no constraints are active, (b) the first constraint is active, (c) the second constraint is active, (d) both constraints are active.

Example Problem:

Let $L(x) = x_1^2 - x_2$ with the constraints $x_2 \leq x_1$ and $x_1 \geq 0$.

We will define our constraint function f to be

$$f(x) = \begin{pmatrix} x_2 - x_1 \\ -x_1 \end{pmatrix} \leq 0$$

. So, our Hamiltonian is

$$H = L + \mu^\top f = x_1^2 - x_2 + \mu_1(x_2 - x_1) + \mu_2(-x_1) = x_1^2 - x_2 + \mu_1(x_2 - x_1) - \mu_2 x_1$$

The necessary conditions are

$$\frac{\partial H}{\partial x_1} = 2x_1 - \mu_1 - \mu_2 = 0 \rightarrow x_1 = \frac{\mu_1 + \mu_2}{2}$$

$$\frac{\partial H}{\partial x_2} = -1 + \mu_1 = 0 \rightarrow \mu_1 = 1$$

We see that

$$x_1 = \frac{1 + \mu_2}{2}$$

We then look at our constraints.

1. Assume no constraints are active ($\mu_1 = \mu_2 = 0$)

We know this cannot be the case because $\mu_1 = 1$ must be held. So, the first constraint needs to be considered.

2. Assume the first constraint is active ($\mu_2 = 0$)

Then, $x_1 = \frac{1}{2}$. We then look at our condition:

$$\frac{\partial H}{\partial \mu_1} = x_1 - x_2 = 0$$

So, $x_1 = x_2$ and therefore our optimal solution is

$$x^* = \left(\frac{1}{2}, \frac{1}{2} \right)$$

which we see has $L(x_1, x_2) = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$.

3. Assume the second constraint is active ($\mu_1 = 0$)

We know this cannot be the case because $\mu_1 = 1$ must be held. So, the first constraint needs to be considered.

4. Assume both constraints are active.

Since both constraints are active, then

$$\frac{\partial H}{\partial \mu_1} = x_1 - x_2 = 0 \rightarrow x_1 = x_2$$

$$\frac{\partial H}{\partial \mu_2} = -x_1 = 0 \rightarrow x_1 = 0$$

So, $x_1 = x_2 = 0$ is the optimal solution here, with $L(0, 0) = 0$.

Knowing our two optimal solutions for the different cases, we see that $(1/2, 1/2)$ produces a more minimal point than $(0, 0)$.

4 Introduction to Optimal Control

4.1 Discrete Optimal Control

Consider the discrete system

$$x_{k+1} = f(x_k, u_k), \quad x_0 \text{ given}$$

with $k = 1, \dots, N$. The optimization problem then is to find

$$\min_{\{u_k\}} \phi(x_N) + \sum_{k=0}^{N-1} L(x_k, u_k) = J(u_0, \dots, u_N)$$

The direct solution is

$$x_k = X_k(u_0, u_1, \dots, u_{k-1})$$

4.1.1 One-Stage Optimal Control

$$x_1 = f(x_0, u_0), \quad x_0 \text{ given}$$

$$\min_{u_0} [\phi(x_1) + L(x_0, u_0)] = \min_{u_0, x_1} J(u_0, x_1)$$

Re-organize the Lagrange multiplier:

$$\bar{J} = J + \lambda_1^\top (f(x_0, u_0) - x_1)$$

which works because $x_1 = f(x_0, u_0)$, so this should evaluate to $\bar{J} = J$ optimally.

$$\bar{J} = \phi(x_1) + L(x_0, u_0) + \lambda_1^\top f(x_0, u_0) - \lambda_1^\top x_1$$

We notice that there is a Hamiltonian there:

$$H(x_0, u_0, \lambda_1) = L(x_0, u_0) + \lambda_1^\top f(x_0, u_0)$$

We then assume that u_0^* is the optimum which produces the corresponding state update x_1^* . Perturbing the system, we end up with

$$dJ = \left. \frac{\partial H}{\partial u} \right|_{x_1^*, u_0^*} du_0$$

Since the necessary condition for optimality is that the perturbation dJ is zero, we see then that it must hold that

$$\frac{\partial H}{\partial u} = 0$$

4.1.2 Multi-Stage Optimal Control

Consider the system

$$x_{k+1} = f_k(x_k, u_k), \quad x_0 \text{ given}$$

with $k = 1, \dots, N$. The performance index (J) is

$$J = \phi(x_N) + \sum_{k=0}^{N-1} L_k(x_k, u_k)$$

Following the same steps as with the one-stage optimal control, we end up with the perturbation

$$dJ = \sum_{k=0}^{N-1} \frac{\partial H_k}{\partial u_k} du_k$$

which should be 0. Which means that

$$\frac{\partial H_k}{\partial u_k} = 0$$

We also have the Lagrange multipliers

$$\lambda_k = \left(\frac{\partial H_k}{\partial x_k} \right)^\top, \quad \lambda_N = \left(\frac{\partial \phi}{\partial x} \right)^\top$$

Note that we define the Hamiltonian to be

$$H_k = L_k + \lambda_{k+1}^\top f_k$$

Our system then has the costate equation of

$$\lambda_k^\top = \frac{\partial L_k}{\partial x_k} + \lambda_{k+1}^\top \frac{\partial f}{\partial x_k}$$

This means our state and costate equations are coupled in such a way that we would call this a Two Point Boundary Value Problem. These are not relatively easy to solve.

4.2 Calculus of Variations

We cannot use this concept for systems in continuous time, so we introduce the concept of calculus of variations. Let there be a performance index

$$J = \int_{x_0}^{x_1} F(x, y, y') dx$$

Why have the derivative y' in the function? Without it, finding the optimum becomes a much simpler problem. However, if we add constraints, then the optimal solution falls apart. We need the derivative of y so that we can guarantee that the boundary conditions are satisfied.

We introduce an arbitrary function $\eta(x)$ which is continuous and differentiable in $[x_0, x_1]$. Suppose then that $y^*(x)$ is the optimal solution. We perturb as such:

$$y(x) = y^*(x) + \epsilon \eta(x)$$

where ϵ is positive but goes to 0 (arbitrarily small). Then the derivative is

$$y'(x) = y^{*'}(x) + \epsilon \eta'(x)$$

Our performance index (depending on ϵ) is

$$J(\epsilon) = \int_{x_0}^{x_1} F(x, y^*(x) + \epsilon \eta(x), y^{*'}(x) + \epsilon \eta'(x)) dx$$

Now, let $\eta(x_0) = \eta(x_1) = 0$. Doing a Taylor expansion, we end up with the equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} = 0$$

which is the condition for optimality for $y^*(x)$. This is called the Euler-Lagrange equation.

Theorem 4.1. Fundamental Lemma of Calculus of Variations

If $\varphi(t)$ is continuous in $[t_0, t_1]$ and if for all $\eta(t) \in C^1$ such that $\eta(t_0) = \eta(t_1) = 0$

$$\int_{t_0}^{t_1} \eta(t) \varphi(t) dt = 0,$$

then it must hold that

$$\varphi(t) \equiv 0.$$

4.3 Optimal Controls with No Constraints

Let our control problem be

$$\arg \min_{u(\cdot)} \int_0^T L(x(t), u(t), t) dt + \phi(x(T))$$

such that

$$\dot{x}(t) = f(x(t), u(t), t)$$

with $T < \infty$ being fixed. We can rewrite L to be

$$L(x, u, t) = L(x, u, t) + \lambda^\top(t)(f(x, u, t) - \dot{x}) = H(x, u, t, \lambda) - \lambda \dot{x}$$

Then our performance index is

$$J = \phi(x(T)) + \int_0^T [H(x, u, t, \lambda) - \lambda \dot{x}] dt$$

which then can be rewritten as

$$J = \int_0^t [H + \dot{\lambda}^\top x] dt + \phi(x(T)) - \lambda^\top(T)x(T) + \lambda^\top(0)x(0)$$

Through this we end up with

$$\dot{\lambda} = - \left(\frac{\partial H}{\partial x} \right)^\top, \quad \lambda(T) = \left(\frac{\partial \phi}{\partial x} \right)^\top$$

which is our Euler-Lagrange equation.

5 Optimal Control in Continuous Time

Through everything we have gone through already, we can set up the simplest problem to have the state equation

$$\dot{x} = f(x, u, t), \quad x_0, T$$

for its dynamics and the performance index

$$J = \phi(x(T)) + \int_0^T L(x, u, t) dt$$

To solve the problem we look at the

1. Hamiltonian:

$$H(x, u, \lambda, t) = L(x, u, t) + \lambda^\top f(x, u, t)$$

2. Optimality Condition (O.C.):

$$\frac{\partial H}{\partial u} = 0$$

3. Euler-Lagrange Equation (E.L.):

$$\dot{\lambda} = - \left(\frac{\partial H}{\partial x} \right)^\top, \quad \lambda(T) = \left(\frac{\partial \phi}{\partial x} \right)^\top$$

4. State Equation (S.E.):

$$\dot{x} = \left(\frac{\partial H}{\partial \lambda} \right)^\top, \quad x(0) = x_0$$

Simple Example:

Consider the scalar linear system

$$\dot{x} = ax + bu$$

with the performance index

$$J = \frac{1}{2} \int_0^T (x^2 + \rho u^2) dt$$

The Hamiltonian is

$$H = \frac{1}{2}x^2 + \frac{\rho}{2}u^2 + \lambda(ax + bu)$$

The optimality condition is

$$\begin{aligned} \frac{\partial H}{\partial u} &= \rho u + b\lambda = 0 \\ u &= -\frac{b}{\rho}\lambda \end{aligned}$$

The Euler-Lagrange equation is

$$\begin{aligned} \dot{\lambda} &= -(x + a\lambda) \\ \lambda(T) &= 0 \end{aligned}$$

Plugging everything back into our state equation:

$$\dot{x} = ax + b \left(-\frac{b}{\rho}\lambda \right)$$

We can set up an equation like so:

$$\begin{bmatrix} \dot{x} \\ \dot{\lambda} \end{bmatrix} = \begin{bmatrix} a & -\frac{b^2}{\rho} \\ -1 & -a \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = M \begin{bmatrix} x \\ \lambda \end{bmatrix}$$

Then, we see that

$$\begin{bmatrix} x(t) \\ \lambda(t) \end{bmatrix} = e^{Mt} \begin{bmatrix} x_0 \\ \lambda(0) \end{bmatrix}$$

Then,

$$\begin{aligned} \begin{bmatrix} x(T) \\ \lambda(T) \end{bmatrix} &= \begin{bmatrix} x(T) \\ 0 \end{bmatrix} = e^{MT} \begin{bmatrix} x_0 \\ \lambda(0) \end{bmatrix} \\ \begin{bmatrix} x_0 \\ \lambda(0) \end{bmatrix} &= e^{-MT} \begin{bmatrix} x(T) \\ 0 \end{bmatrix} \end{aligned}$$

Our unknowns are $\lambda(0)$ and $x(T)$ which we will have to find. Doing this will let us then solve for $x(t)$ and $\lambda(t)$ at any time t . This is the hardest part of solving an optimal control problem.

5.1 O.C. with (Some) Specified States at a Final Time

Let us consider the problem where we have the dynamics

$$\dot{x} = f(x, u, t)$$

$$\dim(x) = n$$

with

$$(x_1(T), \dots, x_p(T)), \quad p \leq n$$

prescribed. When we go through our optimal control problem and look at the perturbations to see how we can find the optimum, we end up with

$$\delta J = \int_0^T \frac{\partial H}{\partial u} \delta u dt = 0$$

Normally, since δu is arbitrary due to us not having specified final states, we can just say that $\frac{\partial H}{\partial u} = 0$. However, δu might affect the end states, so it can no longer be arbitrary. Therefore, we can no longer have that optimality condition. We must compute the influence of δu on $\delta x_i(T)$ ($i = 1, \dots, p$). We specify that our overall Hamiltonian for the problem (finding extremum of performance index J) is

$$H^{(J)} = L + (\lambda^{(J)})^\top f$$

Our Lagrange multipliers $\lambda^{(J)}$ for this problem. We then want to find how our perturbations affect the final states. We look at:

$$\bar{x}_i(T) = x_i(T) + \int_0^T (\lambda^{(i)})^\top (f(x, u, t) - \dot{x}) dt$$

Where $\lambda^{(i)}$ is completely free. We define the Hamiltonian for this subproblem to be

$$H^{(i)} = (\lambda^{(i)})^\top f$$

Our choice of $\lambda^{(i)}$ satisfies its Euler-Lagrange equation:

$$\dot{\lambda}^{(i)} = - \left(\frac{\partial H^{(i)}}{\partial x} \right)^\top$$

$$\lambda^{(i)}(T) = \frac{\partial x_i(T)}{\partial x} = e_i$$

where e_i is the i -th column of the identity matrix. We end up with the necessary condition

$$\delta x_i(T) = \int_0^T (\lambda^{(i)})^\top \frac{\partial f}{\partial u} \delta u dt = \int_0^T \frac{\partial H^{(i)}}{\partial u} \delta u dt = 0$$

The trick is then to let our optimal δu be

$$\delta u^*(\nu_1, \dots, \nu_p, k) = k \left[\frac{\partial H^{(J)}}{\partial u} + \sum_{i=1}^p \left(\frac{\partial H^{(i)}}{\partial u} \right)^\top \nu_i \right]$$

The new problem is now to find ν such that the final state constraints are satisfied.

$$\delta x_i(T) = k \int_0^T (\lambda^{(i)})^\top \frac{\partial f}{\partial u} \left[\frac{\partial H^{(J)}}{\partial u} + \sum_{i=1}^p \left(\frac{\partial f}{\partial u} \right)^\top \lambda^{(i)} \nu_i \right] dt$$

We set $\delta x_i(T) = 0$, which is our objective. Therefore, k is irrelevant. We say that

$$q_i = \int_0^T (\lambda^{(i)})^\top \frac{\partial f}{\partial u} \left(\frac{\partial H^{(J)}}{\partial u} \right)^\top dt$$

$$Q_{ii} = \int_0^T (\lambda^{(i)})^\top \left(\frac{\partial f}{\partial u} \right) \left(\frac{\partial f}{\partial u} \right)^\top \lambda^{(i)} dt$$

So that our problem becomes

$$\delta x_i(T) = q_i + \sum_{i=1}^p Q_{ii} \nu_i = q_i + Q_i \nu = 0$$

Therefore, we can solve for ν :

$$\nu = -Q^{-1}q$$

Note that Q must be symmetric and invertible. We see that Q is also the Reachability Gramian. Basically, our problem hinges on Q not having a zero eigenvalue. If $\det Q = 0$, then there is no solution to the problem. In other words, if our system is not reachable, we are unable to reach those final state conditions. If Q satisfies these conditions, then $\delta x_i(T) = 0$ is guaranteed.

From this, we get

$$\delta J = \int_0^T \|\delta u^*\|^2 dt = 0$$

so our condition is now that $\delta u^*(\nu^*, k) = 0$

We can produce a super Hamiltonian for the overall problem:

$$\begin{aligned} H^{\text{super}} &= H^{(J)} + \sum_{i=1}^p \nu_i H^{(i)} \\ &= L + (\lambda^{(J)})^\top f + \sum_{i=1}^p (\lambda^{(i)})^\top f \nu_i \\ &= L + \left[\lambda^{(J)} + \sum_{i=1}^p \nu_i \lambda^{(i)} \right]^\top f \end{aligned}$$

So we can say we also have a super Langrange multiplier

$$\lambda^{\text{super}} = \lambda^{(J)} + \sum_{i=1}^p \nu_i \lambda^{(i)}$$

so that

$$H^{\text{super}} = L + \lambda^{\text{super}} f$$

and then our optimality condition boils down to

$$\frac{\partial H^{\text{super}}}{\partial u} = 0$$

Example:

$$\dot{x} = x + u, \quad x(0) = 0 \rightarrow x(1) = 1$$

Our goal is to find the minimal energy

$$J = \frac{1}{2} \int_0^T u^2 dt$$

The Hamiltonian is

$$H = \frac{1}{2} u^2 + \lambda(x + u)$$

The optimality condition is

$$\frac{\partial H}{\partial u} = 0 \rightarrow u = -\lambda$$

The Euler-Lagrange equation says

$$\dot{\lambda} = -\lambda$$

Now, how do we find the final λ ? Originally, the final λ is

$$\lambda(T) = \left(\frac{\partial \phi}{\partial x} \right)^\top$$

However, in this case, it also depends on x . So, we create an expression:

$$\psi = \phi + \nu x$$

and then say that

$$\lambda(T) = \left(\frac{\partial \psi}{\partial x} \right)^\top$$

In this problem, $\phi = 0$, so

$$\lambda(T) = \frac{\partial}{\partial x}(\nu x) = \nu$$

We see then that

$$\lambda(t) = e^{T-t} \nu$$

We then plug this back into our state equation:

$$\dot{x} = x + u = x - e^{T-t} \nu$$

which produces

$$x(t) = e^t x(0) - \int_0^t e^{t-\tau} e^{T-\tau} \nu d\tau = -\nu e^T \sinh(t)$$

Then,

$$\begin{aligned} x(T) &= 1 = \nu e^T \sinh(T) \\ \nu &= -\frac{e^{-T}}{\sinh(T)} \end{aligned}$$

So,

$$u^*(t) = -\lambda(t) = -e^{T-t} \nu = e^T e^{-t} \frac{e^{-T}}{\sinh(T)} = \frac{e^{-t}}{\sinh(T)}$$